

Nourhene Ferchichi

Ariana, Tunisia | nourhene.ferchichi2001@gmail.com | +216 95 807 707 | github.com/Nourhene123
| linkedin.com/in/nourhene-ferchichi | portfolio-nourheneferchichi.vercel.app

Summary

Software engineer specializing in full-stack and AI development. Experienced with SaaS platforms, microservices architectures, and LLM-based AI agents, from design to production deployment on AWS. Proficient with Next.js, NestJS, FastAPI, Kubernetes, and multi-agent systems.

Experience

Full-Stack & AI Engineer, Graduation Internship — SmartOvate (Remote) Jan 2026 – Jun 2026

- Subul: an AI-agent learning platform that personalizes IT certification roadmaps (AI, Cloud, Cybersecurity, IoT). Microservices architecture with 6 services, RabbitMQ messaging, Redis cache, one PostgreSQL database per service, Turborepo monorepo.
- Built an 8-agent AI pipeline: a ReAct and LangGraph orchestrator coordinating 5 tools (learner diagnostics, adaptive question generation, RAG-based roadmap generation, self-critique, conversational negotiation), using Azure OpenAI GPT-4o-mini and Azure AI Search.
- DevOps: GitHub Actions CI/CD (Jest tests, Docker, ECR, Helm) on Amazon EKS, Multi-AZ (VPC, NAT Gateway, ELB), Grafana monitoring.
- Stack: Next.js, NestJS, FastAPI, LangGraph, Turborepo, PostgreSQL, Redis, RabbitMQ, Docker, AWS (EKS, ECR, RDS, ALB), Terraform, Helm.

Software Engineering Intern — NST Group — Tunis, Ariana (on-site) Jun 2025 – Aug 2025

- University management platform (Tunisian LMD system) built with Spring Boot 3.5 and Angular 16: role-based access control (RBAC) via Spring Security, 7-step guided enrollment workflow, coefficient-weighted grade calculation, real-time dashboards per role (admin, teacher, student). Improved administrative efficiency by 40 percent.
- Stack: Spring Boot, Spring Security, Angular, PostgreSQL, JWT, Chart.js, GitLab.

Full-Stack Developer Intern, Graduation Internship — Coffee&Brackets — Tunis / Bizerte (on-site) Feb 2023 – Jun 2023

- Coworking space reservation platform (React, NestJS, PostgreSQL): user, resource, and billing management, interactive availability calendar, JWT authentication, automated admin dashboard that cut management time by 50 percent. Handled 500+ bookings.
- Stack: React, NestJS, PostgreSQL, TypeScript, JWT, REST API.

Projects

- **MedFlow**, medical clinic management SaaS (in production): multi-tenant, 5 RBAC roles (Admin, Clinic Manager, Receptionist, Doctor, Patient), real-time appointments (WebSockets, drag-and-drop calendar), Stripe payments and PDF invoice/prescription generation, AI diagnostic chat, Amazon EKS deployment.
Stack: Next.js 15, React 19, Django 5 / DRF, PostgreSQL, Redis, WebSockets, Stripe, Docker, Kubernetes, JWT.
- **SmartRecruitAI**, intelligent CV and job-offer matching: Airflow ETL pipeline, NLP extraction (spaCy, PyMuPDF), hybrid semantic matching, RAG-based explainability engine. Fine-tuned all-mpnet-base-v2 on 10,000+ real CVs (loss 0.018 over 2 epochs).
Stack: Django, Angular, PostgreSQL, Airflow, Sentence-Transformers, spaCy, RAG, NLP.
- **HR Automation Platform**, AI-assisted HR platform: NLP-based CV parsing (Ollama), skill matching with scoring, candidate and employee portals. Automated 80 percent of CV screening.
Stack: .NET, Angular, SQL Server, Ollama, NLP, GitLab, CI/CD.

Education

TEK-UP University — M.Eng. in Software Engineering and Information Systems (graduated)	2024 – 2026
Faculty of Sciences of Bizerte — Professional Master's in Data Science, first year (M1)	2023 – 2024
Faculty of Sciences of Bizerte — Bachelor's in Software Engineering and Information Systems	2020 – 2023
Lycée Mixte de Menzel Bourguiba — High School Diploma, Experimental Sciences	2019 – 2020

Technical Skills

Programming: Java, Python, TypeScript, JavaScript
Full-Stack: Next.js, Angular, NestJS, Django, FastAPI, Spring Boot, Hibernate
Databases: PostgreSQL, MySQL, MongoDB, Redis
Cloud and DevOps: AWS (EKS, ECR, S3, ALB, RDS, Lambda), Docker, Kubernetes, Terraform, Helm, GitHub Actions CI/CD, Grafana
AI: LangChain, LangGraph, RAG, ReAct agents, Azure AI (Foundry, AI Search), OpenAI GPT-4o-mini, NLP, Fine-tuning
Methodology: Agile, Scrum, ClickUp
Languages: English (professional), French (fluent), Arabic (native)

Volunteer Experience

Local Committee Vice President, Outgoing Global Volunteer — AIESEC in Bizerte Feb 2024 – Feb 2025

- Led a team of 20: growth strategy, KPI tracking, monthly meetings and 1-on-1 sessions, international calls for volunteering opportunities abroad. Increased participation by 10 percent.

Agenda Manager, Local Conference — AIESEC in Bizerte Aug 2024 – Oct 2024

- Coordinated a local conference with 90+ participants: sessions, themes, logistics, real-time scheduling, and speaker management.